

Rahul Singh

Machine Learning Engineer | Applied AI, LLM Systems & Production ML

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PROFESSIONAL SUMMARY

Machine Learning Engineer with **4+ years** at Intangles building production telemetry ML and an internal RAG recommender. Reduced false-positive fuel alerts by **15%**; builds LLM tool workflows with deterministic analytics, evidence checks, and replayable evaluation.

TECHNICAL SKILLS

Generative AI / LLMs: RAG, tool orchestration, structured outputs, LoRA / parameter-efficient fine-tuning

ML / NLP: PyTorch, TensorFlow/Keras, Hugging Face Transformers / PEFT, scikit-learn, LightGBM, sentence-transformers

Retrieval: Embeddings, semantic search, pgvector, Maximal Marginal Relevance (MMR) reranking

Backend & Data: Python, SQL, PostgreSQL, RabbitMQ

MLOps & Cloud: Docker, AWS, MLflow, OpenTelemetry, GitHub Actions (CI/CD), LLM evaluation, provider replay, Git

PROFESSIONAL EXPERIENCE

Intangles — Machine Learning Engineer
Pune, India

Apr 2022 – Present

Spec Tag Recommender | Internal vehicle-tag recommendations

- Owned an internal **RAG recommendation tool** that helped teams assign algorithm-specific vehicle tags, reducing manual effort and tagging errors and supporting new hires with limited specification knowledge.
- Retrieved relevant vehicle examples for **AWS Bedrock** LLM inference using sentence-transformer embeddings, a **PyTorch** projection network, and **pgvector**; applied confidence tiers and Maximal Marginal Relevance (MMR) reranking to select supporting context.
- Implemented **RabbitMQ** recommendation and embedding consumers to process requests and update the vector index; added JSON parsing, type coercion, and deterministic rules to check and normalize model outputs.
- Monitored, maintained, and debugged Spec Tag and Fuel Analytics releases; used **OpenTelemetry** traces and **MLflow/S3** artifacts for diagnostics, with SRE handling deployment.

Fuel Analytics | Customer-facing detection and volume estimation

- Owned algorithm development for a customer-facing fuel analytics product, extending **DBSCAN/LOESS** event detection on OBD telemetry.
- Developed refill-volume estimation with recorded results within **±2%** of volumes in customer-entered filling logs.
- Reduced false-positive alerts by **15%** across subsequent releases using simple moving average (SMA) smoothing on fuel-level signals.
- Built **sub-threshold detection** for small thefts and refills missed by existing threshold logic and sensor limitations.

PROFESSIONAL EXPERIENCE (CONTINUED)

EV Battery Coolant Temperature | Team-based inference development

- Prepared irregular OBD/CAN telemetry for coolant-temperature inference using resampling and discharge filtering.
- Built ambient-temperature features for **TensorFlow/Keras LSTM** inference that produced contextual and forecast health states.
- Added data-sufficiency checks and physical-bound validation to screen incomplete inputs and implausible forecasts.
- Implemented output clipping and deterministic fallbacks for missing data and degraded inference paths.

AI / MACHINE LEARNING PROJECTS

FIN-AI | CLI Financial Research Agent

github.com/rahul2-byte/financial-analyst-system

- Built a CLI-first NSE/BSE research agent with deterministic lookup routes, optional typed **Jev** decisions, bounded **Hive GLM-5.3-Flash** tool execution, and nine schema-validated tools.
- Kept financial calculations in deterministic **Python/TA-Lib** while integrating Upstox, YFinance, and TinyFish evidence; added bounded Hive retries and Upstox-to-YFinance fallback on supported paths.
- Implemented fail-closed report checks for cited sources, major-claim references, and numeric fact bindings; invalid drafts receive structured repair or a limited-evidence outcome.
- Built a **150-case** live/replay harness; one Hive run recorded **143/150 terminal successes (95.3%)** with **45.7s p50 / 110.3s p95** latency. Validated **555 tests** at **81% application source coverage**.

LoRA Reproduction & Parameter-Efficient RoBERTa Fine-Tuning Study

github.com/rahul2-byte/lora-reproduction

- Implemented and tested custom **PyTorch** LoRA adapters for RoBERTa query/value projections, including frozen base weights, rank/alpha scaling, adapter dropout, paper-style initialization, safe module injection, and merged inference export.
- Designed a reproducible **27-run** GLUE study: **18 core runs** across MRPC/SST-2 and three seeds, one **PEFT** reference run, and eight rank, scaling, and module-placement ablations.
- On one **RTX 4050** benchmark, reduced trainable parameters from **124.6M to 887K (99.29%)**, task artifacts from **498.7 MB to 3.57 MB (99.28%)**, and PyTorch peak allocated GPU memory by **61.9–64.6%** versus full fine-tuning.
- Achieved **92.81% ± 0.46%** SST-2 validation accuracy versus **93.04% ± 0.24%** for full fine-tuning across three local seeds; validated PEFT forward parity (maximum absolute error **0.0**) and implemented checkpoint/resume recovery, artifact hashing, leakage audits, and discrepancy analysis.

EDUCATION

Diploma in Computer Science and Engineering

Government Polytechnic Kashipur, Uttarakhand, India

Jul 2018 – Sep 2021